Name: Jay Yende
SEC: B Sem: V PRN: 24070521139

UTP Cable Crimping with RJ45

An RJ45 connector is used to terminate a **UTP (Unshielded Twisted Pair) cable** for Ethernet networking.
Crimping means pressing the connector onto the cable using a **crimping tool** so that the internal pins make contact with the wires.

1. Components Used



UTP cable
(Cat5e / Cat6)

RJ45 connector
(8P8C)

Crimping tool
(for RJ45)

2. RJ45 Pinout (T568B)



(T568B order)

1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

3. RJ45 Pinout (T568A)



(T568A order)

1	White/Green
2	Green
3	White/Orange
4	Blue
5	White/Blue
6	Orange
7	White/Brown
8	Brown

1. Strip the cable



- Remove about 2.5 – 3 cm of the outer jacket.
- Do not cut the inner wires.

2. Arrange the wires



- Untwist the pairs and straighten the wires.
- Arrange them in T568B (or T568A) order.

3. Trim the wires



- Cut the wires to an even length (about 1.2 cm / 0.5 inch).

4. Insert into RJ45



- Hold the connector with the clip facing down.
- Insert the wires into the connector until they reach the end.

5. Crimp



- Place the connector in t crimping tool.
- Squeeze firmly until the connector is crimped.

6. Final Result



- A properly crimped RJ45 connector with the wires aligned and secured.

7. Cable Tester (Verification)



- Connect both ends of the crimped cable to a LAN cable tester.
- Check the LED indications.
- If all LEDs are in sequence (1 to 8), the cable is properly crimped.
- If any LED is missing or out of order, re-crimp the connector.

Straight-through Cable (Same Standard)



RJ45 (T568B) RJ45 (T568B)

Important Tips

- Use the same colour-coding standard (T568B or T568A) at both ends for a straight-through cable.
- Keep the wire pairs as twisted as much as possible until the last step.
- Make sure all 8 wires are inserted into the connector.
- Ensure the cable jacket is inside the connector.
- Use a good quality crimping tool.
- Use a crossover cable (T568A on one end and T568B on the other) for direct PC-to-PC connection (if required).